Branch/Class/Batch: AI&DS/T.E/A1
Student Roll no.: A16
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EXPERIMENT NUMBER: 2

AIM: Simulate a wired LAN using networking devices and test connectivity using PING utility

OBJECTIVES

1. To simulate a wired LAN using networking devices.
2. To configure IP addresses on network devices.
3. To establish communication between devices in a LAN.
4. To test network connectivity using the **PING** utility.
5. To understand the basic operation of a wired LAN.

THEORY:

1. Local Area Network (LAN)

A **Local Area Network (LAN)** is a computer network that connects multiple devices within a limited geographical area such as a home, office, school, laboratory, or college campus. It enables users to share data, printers, files, and internet connections efficiently. A **wired LAN** uses Ethernet cables to provide high-speed, reliable, and secure communication between connected devices.

2. Network Switch

A **network switch** is a networking device that connects multiple computers within a LAN. It receives data from one device and forwards it only to the intended destination using the device's **MAC (Media Access Control) address**. This intelligent forwarding reduces unnecessary traffic, minimizes collisions, and improves overall network performance.

3. IP Address

An **IP (Internet Protocol) Address** is a unique numerical address assigned to every device connected to a network. It allows devices to identify and communicate with each other. For successful communication in a LAN, all devices must belong to the same network and have correctly configured **IP addresses** and **subnet masks**.

Example:

- **PC0:** 192.168.1.1
- **PC1:** 192.168.1.2
- **Subnet Mask:** 255.255.255.0

4. PING Utility

PING (Packet Internet Groper) is a network diagnostic utility used to test the connectivity between two devices. It uses the **Internet Control Message Protocol (ICMP)** to send **Echo Request** packets to a destination device and waits for **Echo Reply** packets.

If the destination device responds successfully, the network connection is considered active. If no response is received, it may indicate problems such as incorrect IP configuration, cable faults, network device failure, or connectivity issues.

Syntax:

ping <IP Address>

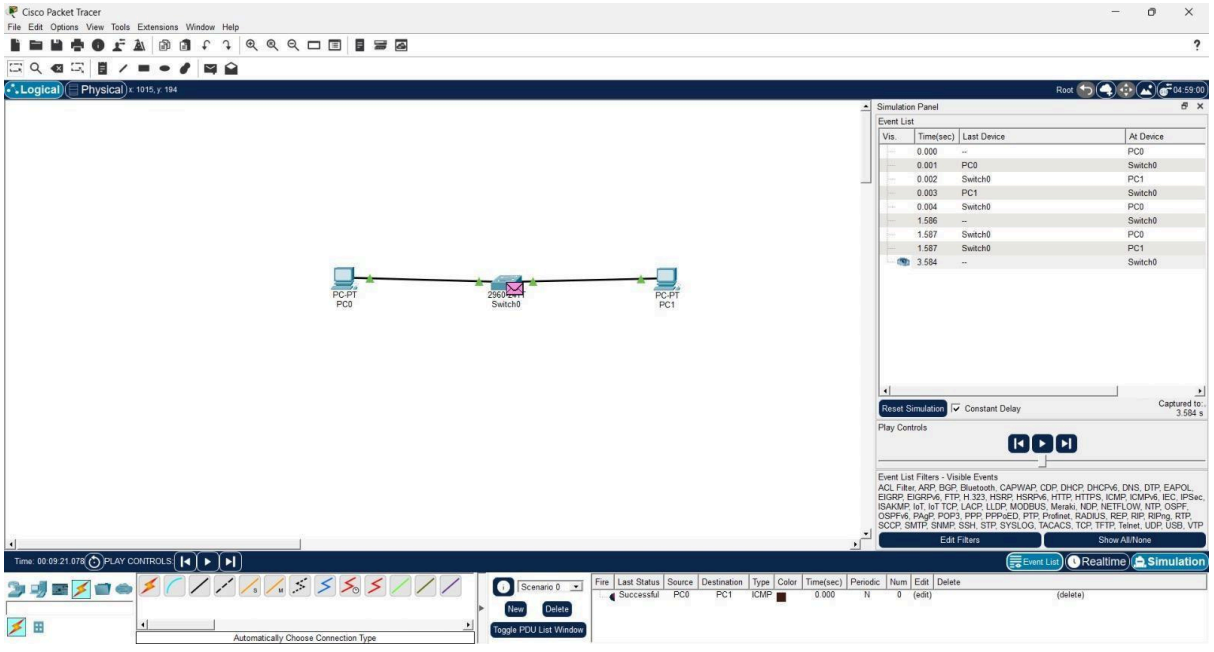
Example:

ping 192.168.1.2

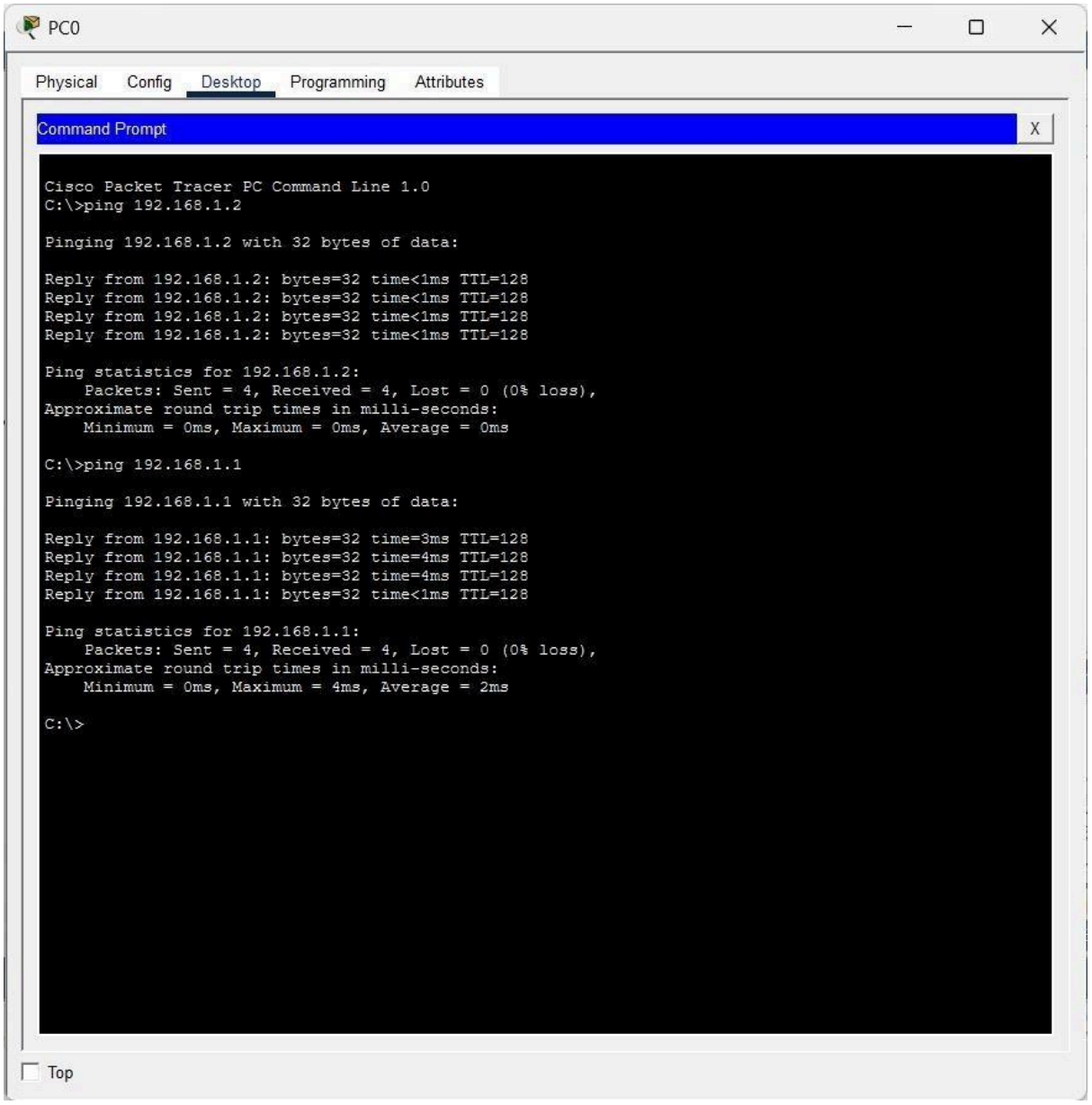
5. Cisco Packet Tracer

Cisco Packet Tracer is a network simulation software developed by Cisco Systems. It allows users to design, configure, simulate, and troubleshoot computer networks without requiring physical networking equipment. It is widely used by students and network professionals to understand networking concepts, practice device configuration, and test network connectivity in a virtual environment.

DIAGRAM:



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RESOURCES REQUIRED

- Cisco Packet Tracer Software
- Personal Computer/Laptop
- Two PC End Devices
- One Cisco 2960 Switch
- Copper Straight-Through Ethernet Cables
- Windows Operating System

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PROCEDURES:

1. Open Cisco Packet Tracer.
2. Drag and place two PCs and one Switch onto the workspace.
3. Connect each PC to the switch using Copper Straight-Through cables.
4. Configure the IP addresses on both PCs:
    - PC0: IP Address: 192.168.1.1, Subnet Mask: 255.255.255.0
    - PC1: IP Address: 192.168.1.2, Subnet Mask: 255.255.255.0
5. Wait until the link lights on the switch turn green, indicate that the connections are active.
6. Open the Command Prompt on PC0 (Desktop → Command Prompt).
7. Execute the following command: ping 192.168.1.2
8. Observe the reply to messages to verify successful communication.
9. Repeat the ping test from PC1 to PC0 using: ping 192.168.1.1
10. Verify that packets are transmitted and received successfully.

RESULT:

A wired Local Area Network (LAN) was successfully created using two personal computers and a Cisco 2960 switch in Cisco Packet Tracer. IP addresses were configured correctly, and the **PING** utility successfully verified communication between both devices. All transmitted packets were received with **0% packet loss**, confirming successful network connectivity.

CONCULSION:

The experiment was performed successfully using Cisco Packet Tracer. A wired LAN was established by connecting two computers through a switch and configuring valid IP addresses. The **PING** utility confirmed successful communication between the devices, demonstrating the basic working of a wired Ethernet network and the importance of correct IP configuration for network connectivity.